

A RESPONSE THEORY FOR THE RIEMANN ZERO GAS: AMPLITUDE CONSERVATION, A THERMAL BRAGG MIRROR, AND THE TWO-SUM INTERFERENCE

UĞUR SEZEN

ABSTRACT. Two companion notes measured how the local statistics of Riemann zeros respond to the prime oscillations of the explicit formula: an amplitude channel $w(\tau)$ and a zero-displacement channel $v(\tau)$, functions of $\tau = \log p/L$, $L = \log \frac{t}{2\pi}$. Here we organize those measurements into a coherent response theory. (i) The two channels obey an *amplitude conservation law*, $\sqrt{w} + (\beta/2)v = 1$ with $\beta/2 = 1.07$, whose coefficient follows from phase bookkeeping alone; the law survives an out-of-sample test on unseen primes, a split-half common-mode test, and a free-exponent measurement ($k = 1.94 \pm 0.18$). (ii) Extending the measurements to $\tau = 0.61$ shows that w *changes sign* at $\tau_0 = 0.40$ — the role-crossover of the companion note — with response phases that jump from 0 to π without rotating; both channels are real everywhere, so the medium is lossless. (iii) A thermal Bragg model explains the crossing: the ideal mirror sits at the Riemann–Siegel fold $\tau = 1/2$; dressing it with the *measured* zero-displacement jitter and the measured indirect-channel plateau moves the zero of the response to 0.406, with unit coherent amplitude ($A = 1.03$); window-to-window jitter differences absorb the previously unexplained fixed- τ scatter. The dephasing scale is the *standing-wave* factor: the empirical characteristic-function ratio gives $s = 4.11 \pm 0.04$ against the Gaussian value 4. (iv) The structure factor $(1 - 2\tau)$ is derived in three lines as the pair count of the folded main sum, vanishing sharply at the fold; pointwise $|Z|^2$ regressions then reveal the complementary *mirror channel*, which follows the shifted-second-moment family $\approx 2(1 - \tau)$ through and beyond the fold, so that the mirror contribution is a *tent* peaked at the fold — the approximate functional equation made audible as an interference channel. (v) The residual variance decomposes as $\sim 5\%$ unstripped tail, 25–45% gap-modulated prime channel, and a core growing at 0.035 per unit L — the CUE rate at one tenth the amplitude.

1. INTRODUCTION

The first companion note [5] measured the joint law of consecutive-zero gaps and between-zero maxima of Hardy’s Z -function and its deviation from CUE characteristic polynomials. The second [6] dissected that deviation into explicit-formula channels: each prime wave $\cos(t \log p)$ enters the amplitudes at full weight and splits between a direct amplitude channel w and a zero-displacement channel v , both collapsing onto one-variable laws in $\tau = \log p/L$. This note asks what *physics* organizes those curves, and answers with five measured structures: a conservation law, a sign-crossing, a thermal Bragg model that computes the crossing’s location, an arithmetic derivation of the structure factor, and the measurement of the complementary mirror channel. Throughout, the style of the companions is kept: no closed forms are fitted; every ingredient is either measured or derived; and every claim is accompanied by the test that tried to kill it.

Data and conventions are those of [5, 6]: eleven amplitude windows spanning $t = 1.2 \times 10^5$ to 2.7×10^{11} (the last from Odlyzko’s table of zeros near the 10^{12} -th [4]), plus six low windows down to $t \approx 1.2 \times 10^3$ for the extended- τ measurements; channels are measured by linear

regression of $\log(\text{unfolded maxima})$ and $\log(\text{unfolded gaps})$ on $\cos(t_n \log p), \sin(t_n \log p)$ at gap midpoints, with own-gap controls for w .

2. THE AMPLITUDE CONSERVATION LAW

The second companion's sum rule — total prime content of unit weight — holds at first order in v . Its violation $g = w + \beta v - 1$ is zero at small τ and grows as $g \approx 1.17 v^2$; and $1.17 \approx \beta^2/4$ with $\beta = 2.14$. The square completes:

$$w = \left(1 - \frac{\beta}{2}v\right)^2 \iff \sqrt{w} + \frac{\beta}{2}v = 1.$$

The coefficient is not fitted from the pair data: since a gap is the derivative of the counting function, the displacement response $U = v/\tau$ is the quadrature partner of the in-phase response w , and $\beta = U(0)/w(0) = 2.01/0.94 = 2.14$ from the small- τ limits alone.

Observation 1 (Amplitude conservation). *Across 44 matched (w, v) pairs ($4 \text{ primes} \times 11 \text{ windows}$, t from 10^5 to 2.7×10^{11}), the free fit gives $\sqrt{w} = 0.991 - 1.031 v$ with residual RMS 0.0091, at the irreducible per-prime scatter floor; constraining the intercept to 1 gives slope $1.068 \approx \beta/2$.*

Three attempts to kill the law failed (Figure 1). *Out of sample*: w for $p = 11, 13$ — never used in deriving the law, and extending τ to 0.26 — follows the line with RMS 0.0148 over 22 new pairs while $\sqrt{w}$ falls from 0.78 to 0.45. *Common mode*: measuring w from even-indexed gaps and v from odd-indexed gaps (independent noise) reproduces the law with $a = 0.998$, $b = 1.051$, RMS 0.019 against 0.018 for same-half pairs. *Free exponent*: fitting $w = (1 - bv)^k$ gives $k = 1.94 \pm 0.18$; among equal-parameter alternatives the intensity-unitarity form $w + cv^2 = 1$ fails by a factor 8 in RMS. What is conserved is the *amplitude*, not the intensity: the prime wave's unit amplitude splits linearly between the transmitted amplitude $\sqrt{w}$ and the absorbed displacement $(\beta/2)v$.

3. THE SIGN CROSSING AND LOSSLESSNESS

Extending the channel measurements to 31 primes over twelve windows (168 points, τ up to 0.61) shows that the amplitude channel does not merely shrink — it *crosses zero* at $\tau_0 = 0.40 \pm 0.02$ and settles on a negative plateau $B = -0.16$: beyond the crossing, peaks are slightly *low* where the prime wave is high. The crossing coincides with the role-crossover $\tau^* \approx 0.40$ found independently by the exhaustion analysis of [6], which is thereby identified: τ^* is the zero of w .

The response phases are diagnostic. Through the crossing the phase of w jumps from $\approx 0^\circ$ to $\approx 180^\circ$ with quadrature components below 3% everywhere; the phase of v stays pinned at 0° . A damped resonance would rotate through $\pm 90^\circ$; a real response that changes sign is the signature of a *lossless* (conservative, Hermitian) medium. The curtain does not absorb the prime waves; it redistributes and reflects them — consistent, at the level of response theory, with a Hermitian generator underneath. This also resolves the conservation law's apparent paradox: w reaches zero not where v would extrapolate the law to closure, but at the sign crossing; the law describes the below-crossing branch.

4. THE THERMAL BRAGG MIRROR

Where should the crossing be? The Riemann–Siegel main sum contains only $n \leq N = \sqrt{t/2\pi}$, i.e. $\log n \leq L/2$: the amplitude has a sharp spectral fold at $\tau = 1/2$, and a wave

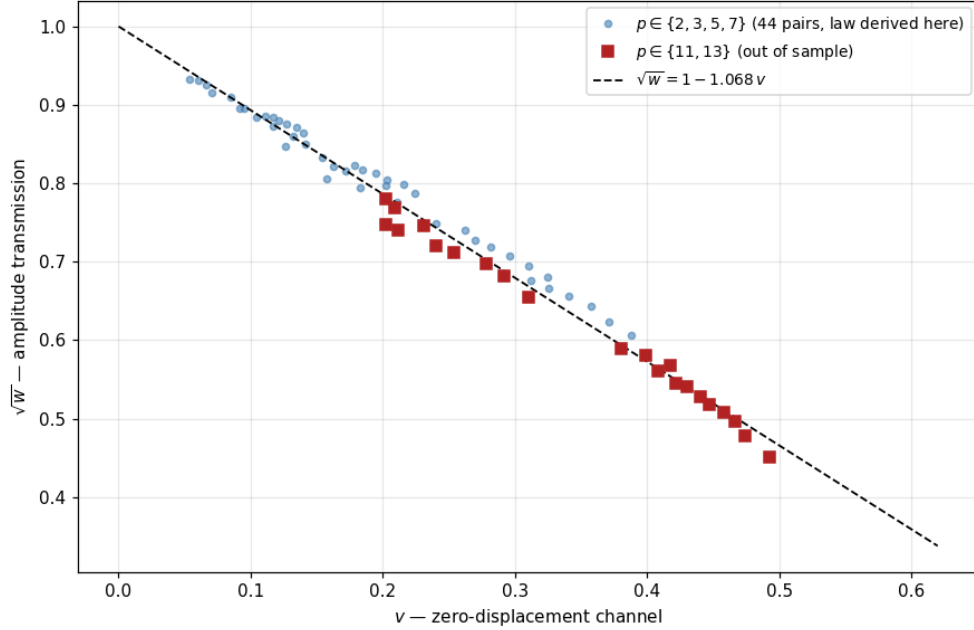


FIGURE 1. The conservation law. Blue: the 44 pairs from which the law was extracted. Red: the 22 out-of-sample pairs ($p = 11, 13$).

whose period matches two lattice spacings ($1/\tau = 2$) is precisely the Bragg condition of the zero lattice. The ideal mirror sits at $\tau = 1/2$; the measured crossing sits at 0.40. The shift is thermal.

Model: $w(\tau) = A(1 - 2\tau)e^{-c\tau^2}[\tau < 1/2] + B$, where $(1 - 2\tau)$ is the ideal structure factor (derived in Section 5), B is the indirect (zero-mediated) channel measured as the beyond-fold plateau, and the Gaussian factor is Debye–Waller dephasing by the zeros’ displacement jitter. The jitter is *measured*: comparing gap midpoints with the Riemann–von Mangoldt flow gives σ_u per window; in unfolded units it grows from 0.198 to 0.272 across our windows — the crystal warms logarithmically, as the classical $S(t)$ variance suggests — giving $c = (L\sigma_u)^2/2 = 0.77\text{--}1.46$, mean 1.17.

Observation 2 (The crossing is computed). *With c fixed from the measured jitter and B fixed from the measured plateau, the single remaining parameter fits to $A = 1.03 \approx 1$ (unit coherent amplitude), and the model’s zero lands at $\tau = 0.406$, on top of the measured 0.40 (Figure 2).*

Two refinements. First, giving each window its own measured $c(L)$ collapses the previously unexplained fixed- τ , window-to-window scatter of the channel measurements from ± 0.04 to ± 0.003 : the “secondary variable” of the companion note is the window temperature. Second, the *scale* of the dephasing is fixed by standing-wave physics: at a Bragg mirror the intracavity field has period $2k$, so dephasing involves e^{2iku} , not e^{iku} . Measured empirically — as the ratio of log characteristic functions of the displacement series, with no Gaussian assumption — this gives

$$s_{\text{eff}} = \frac{\ln |\langle e^{2iku} \rangle|}{\ln |\langle e^{iku} \rangle|} = 4.11 \pm 0.04,$$

the Gaussian value 4 dressed by the measured sub-Gaussian displacement statistics (excess kurtosis -0.3 to -0.5 — level repulsion suppressing the tails). Replacing the Gaussian factor by the empirical $|\langle e^{2iku} \rangle|$ halves the model residual (RMS $0.074 \rightarrow 0.052 \rightarrow 0.029$ for no damping, naive, and standing-wave damping respectively). The remaining excess over the per-prime floor, and the interpolation between the naive and standing-wave crossings (0.408 vs 0.368 , measured 0.40), we attribute to partial reflection below the Bragg condition — an open refinement.

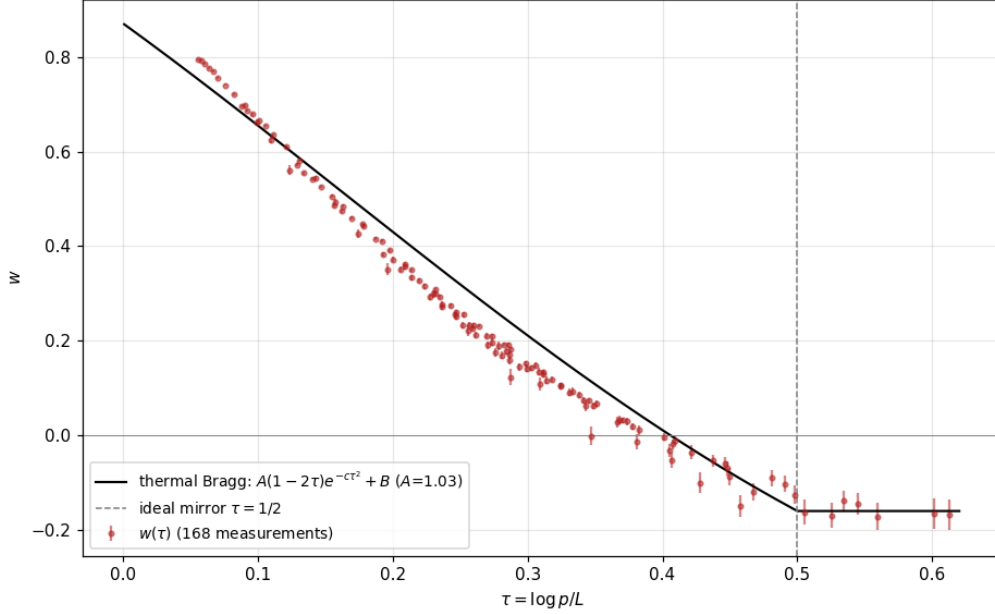


FIGURE 2. The sign crossing and the thermal Bragg model with c from the measured jitter, B from the measured plateau, and $A = 1.03$ the only fitted number.

5. THE STRUCTURE FACTOR, DERIVED

Derivation 3. Write $Z = 2 \sum_{n \leq N} n^{-1/2} \cos \phi_n$, $\phi_n = \theta(t) - t \log n$. In $|Z|^2$ the frequency $\log p$ is produced exactly by the pairs (m, mp) with $mp \leq N$, each contributing weight $4(m \cdot mp)^{-1/2}$, while the DC part is $2 \sum_{n \leq N} n^{-1}$. Hence the relative modulation is

$$R_{\text{direct}}(p) = 2p^{-1/2} \frac{S_1(N/p)}{S_1(N)}, \quad S_1(x) = \sum_{m \leq x} \frac{1}{m},$$

which is $p^{-1/2} \cdot 2(1 - 2\tau)$ in the continuum limit and vanishes identically for $p > N$.

The structure factor $(1 - 2\tau)$ of the Bragg model is therefore not a guess: it is the pair-counting arithmetic of the folded sum, with a sharp horizon at the fold.

6. THE MIRROR CHANNEL: A TENT AT THE FOLD

Testing the derivation pointwise — regressing $|Z(t)|^2$ on the prime waves over fine grids (1.3×10^6 points per window, two windows) — produced a surprise: the measured modulation

does *not* vanish at the fold. It follows, through and beyond the fold, the shifted-second-moment family

$$R(p) \approx 2p^{-1/2} \frac{L - \log p + c_0}{L + c_0} \approx p^{-1/2} \cdot 2(1 - \tau),$$

dying only at the full horizon $p \approx t/2\pi$ (cf. the classical mean-value theorems for $\int |\zeta|^2(m/n)^{it}$ [1]). The difference

$$R_{\text{mirror}} = R_{\text{measured}} - R_{\text{direct}}$$

is a *tent*: it rises as 2τ below the fold, peaks at the fold with value 1, and descends as $2(1 - \tau)$ beyond (Figure 3). Its mechanism is the chirp cross-terms $\cos(2\theta(t) - t \log nm)$ — precisely the interference between the two mirror-image sums of the approximate functional equation — which lock onto frequency $\log p$ at the stationary-phase products $nm \approx (t/2\pi)/p$. Beyond the fold, where no direct pairs exist, everything one hears is the mirrored sum: the functional equation of ζ , measured as an interference channel. The parameter-free tent matches the two-window measurements at the 10–15% level; the exact normalization awaits the true main-term constants (the free fit prefers $c_0 \approx 1.1$ over the naive $2\gamma - 1$), and measurements at $\tau \gtrsim 0.7$ are contaminated by the zero-spacing comb band and are excluded.

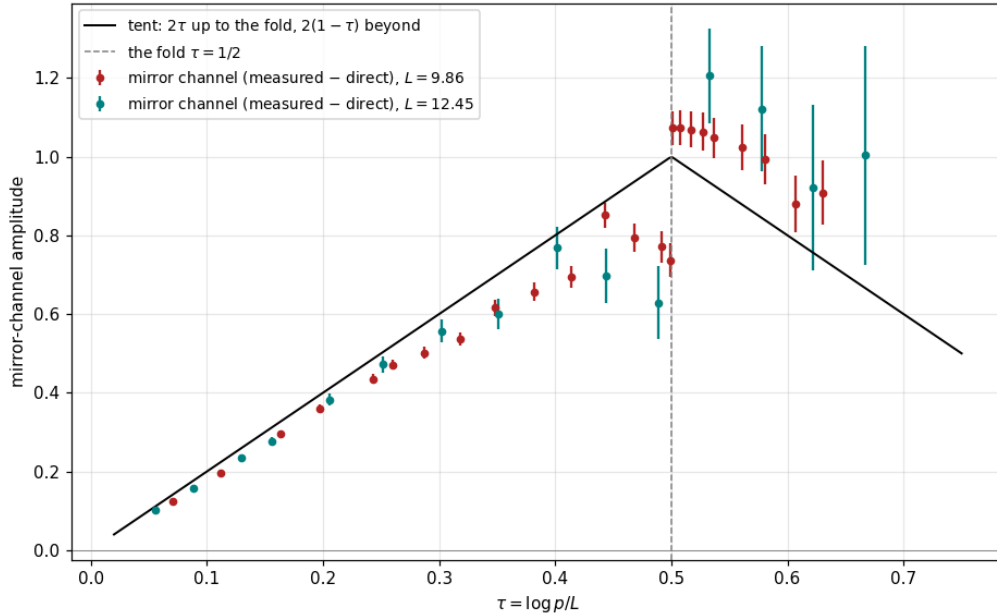


FIGURE 3. The mirror channel at two windows against the parameter-free tent. The direct channel dies at the fold; the mirror channel peaks there.

7. THE RESIDUAL VARIANCE, DECOMPOSED

The companion note’s residual amplitude variance $V_{\text{res}}(L) = 0.163 \rightarrow 0.244$ (after stripping all $p^k \leq 300$) decomposes as follows. The unstripped tail, computed from the measured $w(\tau)$ curve including its negative plateau, contributes only $\sim 4.5\%$. Re-stripping with gap-interaction terms (the transmission is gap-dependent) removes, placebo-corrected, a further $0.074 \rightarrow 0.062$ — 45% to 25% of the residual — and lifts the stripped core correlation to 0.984. What remains grows linearly at 0.035 per unit L : within errors the *CUE amplitude-variance*

rate (0.036 per unit N), at one tenth the magnitude. The zero gas carries an intrinsic noise component that beats at the random-matrix tempo but at a whisper.

8. SYNTHESIS AND OPEN PROBLEMS

The measured response theory of the zero gas now reads: each prime wave enters at unit amplitude and is split, conserving amplitude, between transmission and zero displacement ($\sqrt{w} + (\beta/2)v = 1$); the medium is lossless, with a Bragg mirror at the Riemann–Siegel fold whose observed position is shifted to 0.40 by the measured, logarithmically growing thermal jitter, with standing-wave dephasing of measured scale 4.11; the ideal structure factor is the fold’s pair-counting arithmetic; beyond the fold the response is carried entirely by the functional equation’s mirror sum, whose interference channel is a tent peaked at the fold; and the last residue of randomness beats at the CUE rate at one tenth amplitude.

Open problems: (1) the exact tent normalization from the classical shifted-moment main terms, including the χ -side; (2) the mapping of the pointwise mirror channel to the max-channel plateau $B = -0.16$ (sign and size); (3) partial-reflection dressing $r(\tau)$ below the Bragg condition; (4) a derivation of β ; (5) the distribution (not just variance) of the intrinsic core — is it GUE?; (6) universality: does the same response theory, with the same constants, hold for Dirichlet L -functions?

REPRODUCIBILITY

All measurements are scripts 62–76 in <https://github.com/azizran/deney> (directory `qm_riemann/`), runnable from the stored datasets; the dated research log records the failed intermediate models alongside the results.

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INDEPENDENT RESEARCHER, ISTANBUL, TURKEY

Email address: ugursezen@gmail.com